



Delirium

Full Review: Feb 2025 By [Juebin Huang](#), MD, PhD, Department of Neurology, University of Mississippi Medical Center | Peer reviewed by [Michael C. Levin](#), MD, College of Medicine, University of Saskatchewan
Last updated: Feb 2025

Delirium is an acute, transient, usually reversible, fluctuating disturbance in attention, cognition, and consciousness level. Causes include almost any medical condition (especially when patients are in stressful environments such as hospital settings) or drug exposure. Diagnosis is clinical; laboratory and imaging testing can help identify the cause. Treatment is correction of the cause and supportive measures.

(See also [Overview of Delirium and Dementia](#).)

Delirium may occur at any age but is more common among older adults. At least 10% of older patients (> 65 years) who are admitted to the hospital have delirium ([1](#)); 15 to 50% experience delirium at some time during hospitalization, and it is particularly common after surgery and in patients admitted to an intensive care unit (ICU). Delirium is also common among nursing home residents. When delirium occurs in younger people, it is usually due to use of a drug (recreational drug or medication) or a life-threatening systemic disorder.

Delirium is sometimes called acute confusional state or toxic-metabolic encephalopathy.

Delirium and [dementia](#) are distinct cognitive disorders but are sometimes difficult to distinguish. The following characteristics help distinguish them (see table [Differences Between Delirium and Dementia](#)).

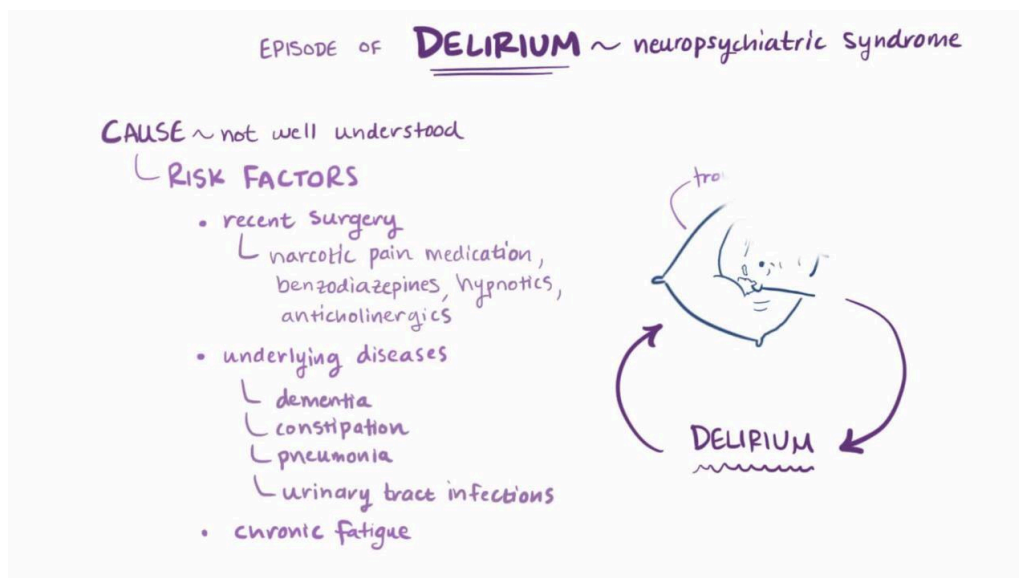
- **Delirium** affects mainly attention, is typically caused by acute illness or drug toxicity (sometimes life threatening), and is often reversible.
- **Dementia** affects mainly memory, is typically caused by anatomic changes in the brain, has slower onset, and is generally irreversible.

Delirium often develops in patients with dementia and is called delirium superimposed on dementia (DSD). DSD can occur in up to 49% of patients with dementia during hospitalization ([2](#)). Also, patients with delirium have a higher risk of developing dementia.

General references

Overview of Delirium

VIDEO



1. [Inouye SK, Westendorp RG, Saczynski JS](#): Delirium in elderly people. *Lancet* 383(9920):911–922, 2014. doi:10.1016/S0140-6736(13)60688-1
2. [Fong TG, Inouye SK](#): The inter-relationship between delirium and dementia: The importance of delirium prevention. *Nat Rev Neurol* 18 (10):579–596, 2022. doi: 10.1038/s41582-022-00698-7

Etiology of Delirium

The **most common causes** of delirium are the following:

- Medications, particularly anticholinergics and opioids or other medications and substances with psychoactive effects
- Dehydration
- Infection

Multiple other conditions can cause delirium (see table [Causes of Delirium](#)). In some cases, no cause can be identified.

Predisposing factors include brain disorders (eg, [dementia](#), [stroke](#), [Parkinson disease](#)), advanced age, sensory impairment (eg, impaired vision or hearing), alcohol intoxication, and multiple coexisting disorders.

Precipitating factors include use of medications (particularly ≥ 3 new medications), infection (eg, urinary tract infection, viral disease), dehydration, shock, hypoxia, anemia, immobility, undernutrition, use of bladder catheters (whether urinary retention is present or not), hospitalization, pain, sleep deprivation, decreased sensory stimuli at night, and emotional stress. Unrecognized liver or kidney failure may cause delirium by impairing the metabolism and reducing the clearance of a previously well-tolerated medication. Recent exposure to anesthesia also increases risk, especially if exposure is

prolonged and if anticholinergics are given during surgery. After surgery, pain and the use of opioid analgesics can contribute to delirium.

For older patients in an ICU, risk of delirium (ICU psychosis) is particularly high. [Nonconvulsive status epilepticus](#) is an underrecognized cause of altered mental status in ICU patients that should be considered.

TABLE

Causes of Delirium

Category	Examples
Neurologic causes	
Cerebrovascular disorders	Hemorrhagic stroke , ischemic stroke , transient ischemia attack
Migraine	Confusional migraine*
Inflammation or infection	Acute demyelinating encephalomyelitis , brain abscess , CNS vasculitis, autoimmune encephalopathy, encephalitis , meningitis , meningoencephalitis
Seizure disorders	Nonconvulsive status epilepticus, postictal state
Trauma	Subdural hematoma , traumatic brain injury
Tumor	Meningeal carcinomatosis, primary or metastatic brain tumor
Nonneurologic causes	
Medications and recreational drugs (numerous)	Anticholinergics, antiemetics, antihistamines (eg, diphenhydramine), antihypertensives, some antimicrobials, antipsychotics, antispasmodics, benzodiazepines, cardiovascular medications (often beta-blockers), cimetidine , corticosteroids, digoxin , dopamine agonists, hypnotics, muscle relaxants, NSAIDs, opioids, recreational drugs, sedatives, tricyclic antidepressants

Category	Examples
Endocrine disorders	Adrenal insufficiency , pituitary insufficiency , Cushing syndrome , hyperparathyroidism , hyperthyroidism , hypothyroidism
Hematologic disorders	Hyperviscosity syndrome, leukemic blast cell crisis, polycythemia , thrombocytosis
Infections	Pneumonia , sepsis , systemic infections, UTIs
Injuries	Burns , electrical injuries , fat embolism, heatstroke , hypothermia
Metabolic disorders	Acid-base disturbances , fluid and electrolyte abnormalities (eg, dehydration , hypercalcemia , hyponatremia , hypocalcemia , hyponatremia , hypomagnesemia), hepatic or uremic encephalopathy, hyperosmolality, hyperglycemia , hypoglycemia , hypoxia, Wernicke encephalopathy
Vascular or circulatory disorders	Anemia , cardiac arrhythmias , heart failure , hypoperfusion states, shock
Vitamin deficiency	Thiamin deficiency , vitamin B12 deficiency
Withdrawal syndromes	Alcohol , barbiturates , benzodiazepines , opioids
Other causes	Change of environment, fever , fecal impaction, hypertensive encephalopathy, liver failure , long stays in an ICU, mental disorders, postoperative states, sensory deprivation, sleep deprivation, hyperthermia, toxins that affect the CNS, urinary retention

* Confusional migraine is characterized by confusion, agitation or lethargy, dysphasia, amnesia, and absence of headache.

CNS = central nervous system; ICU = intensive care unit; NSAIDs = nonsteroidal anti-inflammatory drugs; UTIs = urinary tract infections.

Pathophysiology of Delirium

Mechanisms are not fully understood but may involve

- Reversible impairment of cerebral oxidative metabolism
- Multiple neurotransmitter abnormalities, especially cholinergic deficiency
- Generation of inflammatory markers, including C-reactive protein, interleukin-1 beta and 6, and tumor necrosis factor-alpha

Stress of any kind upregulates sympathetic tone and downregulates parasympathetic tone, impairing cholinergic function and thus contributing to delirium. Older adults are particularly vulnerable to reduced cholinergic transmission, increasing their risk of delirium.

Regardless of the cause, the cerebral hemispheres and arousal mechanisms of the thalamus and brain stem reticular activating system become impaired.

Symptoms and Signs of Delirium

Delirium is characterized primarily by

- Difficulty focusing, maintaining, or shifting attention (inattention)

Consciousness level fluctuates; patients are disoriented to time and sometimes place or person. They may have hallucinations, delusions, and paranoia. Confusion regarding day-to-day events and daily routines is common, as are changes in personality and affect. Thinking becomes disorganized, and speech is often disordered, with prominent slurring, rapidity, neologisms, aphasic errors, or chaotic patterns.

Symptoms fluctuate over minutes to hours; they may lessen during the day and worsen at night.

Other symptoms of delirium may include inappropriate behavior and fearfulness. Patients may become irritable, agitated, hyperactive, and hyperalert; conversely, they may become quiet, withdrawn, and lethargic. Very old people with delirium tend to become quiet and withdrawn—changes that may be mistaken for depression. Some patients alternate between agitation and withdrawal.

Usually, patterns of sleeping and eating are grossly distorted.

Because of the many cognitive disturbances, insight is poor, and judgment is impaired.

Diagnosis of Delirium

- Mental status examination
- Standard diagnostic criteria to confirm delirium
- Thorough history
- Directed physical examination and selective testing to determine cause

Delirium, particularly in older patients, is often overlooked by clinicians. Clinicians should consider delirium (and dementia) in any older patient who presents with impairment in memory or attention.

Mental status examination

Patients with any sign of cognitive impairment require a formal [mental status examination](#).

Attention is assessed first. Simple tests include immediate repetition of the names of 3 objects, digit span (ability to repeat 7 digits forward and 5 backward), and naming the days of the week forward and backward. Inattention (patient does not register directions or other information) must be distinguished from poor short-term memory (patient registers information but rapidly forgets it). Further cognitive testing is futile for patients who cannot register information.

After initial assessment, standard diagnostic criteria, such as those identified in the *Diagnostic and Statistical Manual of Mental Disorders*, 5th Edition, Text Revision (DSM-5-TR) ([1](#)) or the Confusion Assessment Method (CAM) ([2](#)), may be used.

The following features are required for diagnosis of delirium using DSM-5-TR criteria:

- Disturbance in attention (eg, difficulty focusing or following what is said) and awareness (ie, reduced orientation to the environment)
- The disturbance develops over a short period of time (over hours to days) and tends to fluctuate during the day.
- Acute change in cognition (eg, deficits of memory, language, perception, thinking)

In addition, there must be evidence from the history, physical examination, and/or laboratory testing suggesting that the disturbance is caused by a medical disorder, a substance (including drugs or toxins), or substance withdrawal.

CAM uses the following criteria:

- An altered level of consciousness (eg, hyperalert, lethargic, stuporous, comatose) or disorganized thinking (eg, rambling, irrelevant conversation, illogical flow of ideas)

History

History is obtained by interviewing family members, caregivers, and friends. It can determine whether the change in mental status is recent and is distinct from any baseline dementia (see table [Differences](#)

Between Delirium and Dementia). The history helps distinguish a mental disorder from delirium. Psychiatric disorders, unlike delirium, almost never cause inattention or fluctuating consciousness, and onset of psychiatric disorders is nearly always subacute.

Sundowning (behavioral deterioration during evening hours), which is common among patients with dementia who are living in institutions, may be difficult to differentiate from delirium; newly symptomatic deterioration should be presumed to be delirium until proved otherwise.

History should also include use of alcohol, recreational or illicit drugs, nutritional supplements (eg, herbal products), and over-the-counter and prescription medications, focusing particularly on drugs with anticholinergic and/or other central nervous system (CNS) effects and on new additions, discontinuations, or changes in dose, including overdosing.

Physical examination

Examination, particularly in patients who are not fully cooperative, should focus on the following:

- Vital signs
- Hydration status
- Potential foci for infection
- Skin and head and neck
- Neurologic examination

Specific findings can suggest a cause, as with the following:

- Fever, meningismus, or Kernig and Brudzinski signs suggest CNS infection.
- Tremor and myoclonus suggest uremia, liver failure, drug intoxication, medication toxicity, or certain electrolyte disorders (eg, hypocalcemia, hypomagnesemia).
- Ophthalmoplegia and ataxia suggest Wernicke encephalopathy.
- Focal neurologic abnormalities (eg, cranial nerve palsies, motor or sensory deficits) or papilledema suggests a structural CNS disorder.
- Scalp or facial lacerations, bruising, swelling, and other signs of head trauma suggest traumatic brain injury.

Testing

Testing usually includes

- CT or MRI of the head
- Tests for suspected infection (eg, complete blood count [CBC], blood cultures, chest radiography, urinalysis)
- Evaluation for hypoxia (pulse oximetry or arterial blood gases)
- Measurement of electrolytes, blood urea nitrogen (BUN), creatinine, plasma glucose, and blood levels of any medications suspected to have toxic effects
- A urine drug screen

If the diagnosis is unclear, further testing may include liver tests, measurement of serum calcium and albumin, thyroid-stimulating hormone (TSH), vitamin B12, erythrocyte sedimentation rate (ESR) and/or C-reactive protein (CRP), antinuclear antibody (ANA), and a test for syphilis (eg, rapid plasma reagin [RPR] or Venereal Disease Research Laboratory [VDRL] test).

If the diagnosis is still unclear, testing may include cerebrospinal fluid (CSF) analysis (particularly to exclude meningitis, encephalitis, or subarachnoid hemorrhage), measurement of serum ammonia, and testing to check for heavy metals.

If nonconvulsive seizure activity, including status epilepticus, is suspected (suggested by subtle motor twitches, automatisms, and a fluctuating pattern of bewilderment and drowsiness), electroencephalography (EEG) monitoring should be done.

Diagnosis references

1. Diagnostic and Statistical Manual of Mental Disorders, 5th edition, Text Revision (DSM-5-TR). American Psychiatric Association Publishing, Washington, DC, 2022. pp 672-674
2. [Inouye SK, van Dyck CH, Alessi CA, Balkin S, Siegel AP, Horwitz RJ](#): Clarifying confusion: the confusion assessment method. A new method for detection of delirium. *Ann Intern Med* 113(12):941–948, 1990. doi:10.7326/0003-4819-113-12-941

Treatment of Delirium

- Treatment of underlying disorder and removal of exacerbating factors
- Supportive care
- Management of agitation

Correcting the cause (eg, treating infection, giving fluids and electrolytes for dehydration) and removing exacerbating factors (eg, stopping medications) may result in resolution of delirium. Nutritional deficiencies (eg, of thiamin or vitamin B12) should be corrected, and good nutrition and hydration should be provided.

Additional management considerations may vary depending on the patient population and clinical scenario (eg, ICU care, post-operative care in older adults) ([1](#), [2](#)).

Management of agitation

General measures

The environment should be stable, quiet, and well-lit and include visual cues to orient the patient (eg, calendar, clocks, family photographs). Frequent reorientation and reassurance by hospital staff or family members may also help. Sensory deficits should be minimized (eg, by replacing hearing-aid batteries, by encouraging patients who need eyeglasses or hearing aids to use them).

Approach to treatment should be interdisciplinary (involving physicians, physical and occupational therapists, nurses, and social workers); it should involve strategies to enhance mobility and range of

motion, treat pain and discomfort, prevent skin breakdown, ameliorate incontinence, and minimize the risk of aspiration.

Agitation may threaten the well-being of the patient, a caregiver, or a staff member. Simplifying medication regimens and avoiding use of IV lines, bladder catheters, and physical restraints (particularly in long-term care settings) as much as possible can help prevent exacerbation of agitation and reduce risk of injury. However, in certain circumstances, physical restraints may be needed to prevent patients from harming themselves or others. Restraints should be applied by a staff member trained in their use; they should be released at least every 2 hours to prevent injury and discontinued as soon as possible. Use of hospital-employed assistants (sitters) as constant observers may help avoid the need for restraints.

Explaining the nature of delirium to family members can help them cope. They should be told that delirium is usually reversible but that cognitive deficits often take weeks or months to abate after resolution of the acute illness.

Medications

Antipsychotic medications are sometimes used to treat severe agitation, but their routine use is not recommended. These medications do not correct the underlying problem and may prolong or exacerbate delirium.

Low-dose [haloperidol](#) as needed may lessen agitation or psychotic symptoms; occasionally, higher doses are necessary.

Second-generation (atypical) antipsychotics (eg, [risperidone](#), [olanzapine](#), [quetiapine](#)) may be preferred in older patients because they have fewer extrapyramidal adverse effects; however, long-term use may cause weight gain and other adverse effects. These medications are typically given orally and not parenterally.

For delirium caused by withdrawal from alcohol or benzodiazepines, intravenous benzodiazepines are the medications of choice. Their onset of action is more rapid (5 minutes after parenteral administration) than antipsychotics. However, benzodiazepines should be avoided for treatment of all other causes of delirium because they can worsen confusion and sedation.

Treatment references

1. [Barr J, Fraser GL, Puntillo K, et al](#): Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med* 41(1):263–306, 2013. doi:10.1097/CCM.0b013e3182783b72
2. [American Geriatrics Society Expert Panel on Postoperative Delirium in Older Adults](#): American Geriatrics Society abstracted clinical practice guideline for postoperative delirium in older adults [reaffirmed 2021]. *J Am Geriatr Soc* 63(1):142–150, 2015. doi:10.1111/jgs.13281

Prognosis for Delirium

Morbidity and mortality rates are high in patients who have delirium and are admitted to the hospital or who develop delirium during hospitalization; 30 to 50 % of hospitalized patients with delirium die within 1 year (1). These rates may be high partly because such patients tend to be older and to have other serious disorders.

Delirium due to certain conditions (eg, hypoglycemia, drug or alcohol intoxication, infection, iatrogenic factors, medication toxicity, electrolyte imbalance) typically resolves rapidly with treatment. However, recovery may be slow (days to even weeks or months), especially in older patients, resulting in longer hospital stays, increased risk and severity of complications, increased costs, and long-term disability. Some patients never fully recover from delirium. For up to 2 years after delirium occurs, risk of cognitive and functional decline, institutionalization, and death is increased. In a meta-analysis, delirium in surgical and nonsurgical patients was significantly associated with cognitive impairment lasting ≥ 3 months after the delirium episode (2).

Prognosis references

1. [McCusker J, Cole M, Dendukuri N, Han L, Belzile E](#): The course of delirium in older medical inpatients: a prospective study. *J Gen Intern Med* 18(9):696–704, 2003. doi:10.1046/j.1525-1497.2003.20602.x
2. [Goldberg TE, Chen C, Wang Y, et al](#): Association of delirium with long-term cognitive decline: A meta-analysis [published correction appears in *JAMA Neurol* 2020 Nov 1;77(11):1452. doi: 10.1001/jamaneurol.2020.3284]. *JAMA Neurol* 77 (11):1373–1381, 2020. doi:10.1001/jamaneurol.2020.2273

Prevention of Delirium

Because delirium greatly worsens prognosis for hospitalized patients, prevention should be emphasized. Hospital staff members should be trained to take measures to maintain orientation, mobility, and cognition and to ensure sleep, good nutrition and hydration, and sufficient pain relief, particularly in older patients. Family members can be encouraged to help with these strategies.

The number and doses of medications should be reduced if possible.

Geriatrics Essentials: Delirium

Delirium is more common among older adults. About 15 to 50% of older patients experience delirium at some time during a hospital stay. The risk of delirium is particularly high for older patients in an ICU (ICU psychosis).

Stress of any kind impairs cholinergic function, thus contributing to delirium. Older adults are particularly vulnerable to reduced cholinergic transmission, increasing their risk of delirium. Anticholinergic medications can contribute.

Delirium is often the first sign of another, sometimes serious, disorder in older adults.

Causes of delirium in older adults often include less severe conditions:

- Dehydration
- A nonneurologic disorder (eg, [urinary tract infection](#), [influenza](#), [thiamin deficiency](#), [vitamin B12 deficiency](#))
- Pain
- Urinary retention or severe constipation
- Sensory deprivation
- Sleep deprivation
- Stress (any type)
- Use of a bladder catheter

Certain age-related changes make older adults more susceptible to developing delirium:

- Increased sensitivity to drugs (particularly sedatives, anticholinergics, and antihistamines)
- Changes in the brain (eg, atrophy, lower levels of acetylcholine)
- The presence of conditions that increase the risk of delirium (eg, stroke, dementia, Parkinson disease, other neurodegenerative disorders, polypharmacy, dehydration, undernutrition, immobility)

The most obvious symptom of delirium, confusion, may be harder to recognize in older adults.

Younger people with delirium may be agitated, but very old people tend to become quiet and withdrawn—changes that may be mistaken for depression. In such cases, recognizing delirium is even harder.

Memory impairment and inattention may also be the initial symptoms of delirium in older adults. Because older patients are more likely to have dementia than delirium, delirium is often overlooked. Clinicians should consider delirium in any older patient who presents with impairment in memory or attention.

Psychosis due to a psychiatric disorder rarely begins during old age. If a psychosis develops in an older adult, it usually indicates delirium or dementia.

Delirium can be a common presenting symptom in older patients with infection.

In older adults, delirium tends to last longer, and recovery may be slow (days to even weeks or months), resulting in longer hospital stays, increased risk and severity of complications, increased costs, and long-term disability. Some patients never fully recover from delirium.

Pearls & Pitfalls

- Consider delirium in any older patient who presents with impairment in memory or attention.

An [interdisciplinary team](#) able to provide multifaceted treatment measures can optimally benefit older patients with delirium who are hospitalized and prevent potential [iatrogenic complications](#) (eg, [undernutrition](#), dehydration, [pressure ulcers](#)). These complications may have serious consequences in older patients.

Key Points

- Delirium, which is common among hospitalized older patients, is often caused by drugs, dehydration, and infections (eg, urinary tract infections), but can have many other causes.
- Consider delirium in older patients with behavioral changes, particularly those presenting with impaired memory or attention.
- History taken from family members, caregivers, and friends and mental status examination are key to recognizing delirium.
- Thoroughly assess patients with delirium for possible neurologic and systemic causes and triggers.
- Thoroughly review the patient's medications, and stop any potentially contributory medications.
- Approximately 30 to 50% of hospitalized patients with delirium die within 1 year.
- Treat the cause of delirium and provide supportive care; manage agitation with general measures and sometimes medications when necessary.

Drug Information for the Topic



Copyright © 2026 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.